
Production Modes are Disentangled Latent Variables in LLM Hidden States

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Abstract

When a reader processes a passage, many signals about *how* the text was produced — who wrote it, whether a human or a language model wrote it, in what register, and even what keyboard layout shaped its typos — coexist in the same string of tokens. We ask whether these *production modes* share a unified representational substrate inside a mid-size language model, and whether they are geometrically disentangled. We probe four very different modes in the last-token hidden states of QWEN2.5-1.5B: 7-way literary authorship, AI-vs-human authorship, formal-vs-casual register, and QWERTY-vs-Dvorak typo patterns. A single linear probe decodes every mode well above chance, with peak accuracies of 79.2%–99.8% across the four modes. The probe directions are pairwise near-orthogonal (all $|\cos| \leq 0.18$), and cross-mode transfer of probe directions is near chance for orthogonal modes (TYPO $\leftrightarrow$ REGISTER at 0.48) while exposing one expected content overlap (REGISTER $\rightarrow$ AI-VS-HUMAN at 0.74, reflecting LLM formality bias). Layer-depth profiles split the four modes into three distinct tiers: content/register modes saturate at shallow layers, authorship emerges gradually across the stack, and surface-typing artifacts become decodable only at the final layers. Production modes are real, multiple, and largely independent latent variables in LLM hidden states — a unification with consequences for authorship privacy, controllable generation, and AI-text detection.

1 Introduction

When you write a passage, a large language model that reads it can often tell *how* you produced it. It can usually tell whether the text was dictated rather than typed, whether another LLM wrote it, and — in principle — which keyboard layout shaped the pattern of your typos. We call each such latent variable a *modes* of text production. The question this paper asks is whether different production modes share a common geometry inside an LLM, or whether each lives in its own corner of the representation space.

Why does this matter? The same axis matters to several research communities that have so far worked separately on it. AI-text detection treats the AI-vs-human axis as a standalone binary classification problem [Mitchell et al., 2023, Verma et al., 2023, Schäfer and Steinebach, 2025]. Authorship attribution and stylometry treat *who wrote it* as another standalone classification problem with its own decades-old methodology [Rivera-Soto et al., 2021, Sarfati et al., 2025]. Interpretability work on linear probes and sparse autoencoders [Alain and Bengio, 2016, Templeton et al., 2024] shows that high-level features can be linearly decoded from hidden states, but tends to study one feature at a time. Nobody has asked: do these various *modes* coexist in the same residual stream, and are they geometrically disentangled?

The gap. A unified picture would carry practical weight. If keyboard layout, dictation status, or AI-vs-human origin are all recoverable from the same hidden state, then any system that runs an LLM over user text leaks more than the text. If *modes* are orthogonal, activation steering along one mode (e.g., “make this less formal”) should not disturb another (“preserve the author’s voice”). And if AI text occupies a separate latent direction from the actual content of a passage, benchmark

contamination by AI generations is detectable in hidden states even when invisible at the surface. None of these claims is testable without first establishing whether *modes* are jointly represented and whether they share or separate directions.

Our approach. We treat *modes* as a unified latent variable and probe four very different ones — 7-way literary authorship, AI-vs-human authorship, formal-vs-casual register, and QWERTY-vs-Dvorak typo patterns — in the last-token hidden states of QWEN2.5-1.5B. For each *modes*, at each layer $L \in \{0, 2, \dots, 28\}$ and each context length $N \in \{32, 64, 128\}$, we fit a linear probe under 5-fold stratified cross-validation. We then ask three geometric questions: (i) how do the probe directions for different *modes* align with each other (cosine similarity)? (ii) does a probe direction trained on one *modes* carry information about another (cross-mode transfer)? (iii) how does each *modes* emerge as a function of network depth?

Preview of results. A single linear probe decodes every *modes* well above chance and well above random-direction baselines, with peak accuracies of **79.2%–99.8%**. The decoded directions are pairwise near-orthogonal — all six pairwise cosines satisfy $|\cos| \leq 0.18$. Cross-mode transfer is near chance for orthogonal modes (TYPO $\leftrightarrow$ REGISTER at 0.48–0.50) but reveals one expected content overlap: the formality direction recovers 74% of AI-vs-human labels, the latent footprint of ChatGPT’s well-documented formality bias. Layer-depth profiles further split the four *modes* into three tiers: content modes (register, AI-vs-human) saturate by layer 2, authorship rises gradually across the entire stack, and surface-typing artifacts become decodable only at the final layers.

Contributions. We make the following contributions:

- We propose the *modes* abstraction — production mode as a unified latent variable — and instantiate it on four diverse axes spanning content, author identity, register, and surface-typing artifact.
- We show that a single linear probe family on QWEN2.5-1.5B last-token hidden states decodes each *modes* at 79%–99% accuracy, establishing decodability as a property shared across very different mode types.
- We characterise the *geometry* of *modes*: pairwise direction cosines ≤ 0.18 and cross-mode probe transfer near chance for independent modes, providing the first quantitative evidence that production modes are largely disentangled in LLM hidden states.
- We identify a three-tier layer-depth taxonomy — shallow content modes, gradual author modes, and late surface modes — that organises a previously unstructured set of probe findings.

Section 2 surveys prior work on AI-text detection, stylometry, and probing. Section 3 describes our model, corpora, probes, and geometric analyses. Section 4 presents quantitative results. Section 5 interprets the geometry and notes limitations. Section 6 summarises and lists follow-up directions.

2 Related Work

Linear probes and emergent representations. Alain and Bengio [2016] established the linear-probe methodology: train a linear classifier on frozen activations at each layer to test which information is linearly decodable. Li et al. [2022] showed the strongest form of the latent-variable claim on Othello-GPT, where a transformer trained only on move sequences develops a probe-decodable and causally intervenable board state. Our work uses linear probes in the same spirit, but to test whether multiple modes of *text production* (not a single task variable) coexist as disentangled directions in the same residual stream.

Stylometry and authorship attribution. A long literature treats authorial style as a latent variable recoverable from surface text. Rivera-Soto et al. [2021] learn universal authorship embeddings that transfer across domains (reviews, fanfiction, Reddit). Most relevant to us, Sarfati et al. [2025] show that literary style in 13 novels is linearly decodable from LLAMA-3.2-1B last-token embeddings, reaches its sharpest separation at intermediate layers, and lies within ~ 16 PCA dimensions of class-mean differences. We extend this template along two axes: we test *multiple* modes within the same embedding cache, and we measure their geometric *relationships* (orthogonality and transfer), which a single-mode study cannot.

AI-text detection. A fast-moving subfield treats AI-vs-human classification as its own problem. Mitchell et al. [2023] introduce DetectGPT, a zero-shot detector based on local curvature of the LM’s output distribution. Verma et al. [2023] and others fit supervised classifiers on hand-crafted or neural features. Schäfer and Steinebach [2025] obtain F1 above 0.90 with 220 hand-crafted

stylistic features, noting that AI text systematically biases towards greater lexical complexity and lower variation. None of these methods asks whether the AI-vs-human axis shares geometry with stylistic or surface-form axes. Our cross-mode transfer experiment is, to our knowledge, the first to quantify the overlap between an AI-text direction and a register direction within the same model.

Feature discovery via sparse autoencoders. Templeton et al. [2024] train sparse autoencoders on Claude 3 Sonnet and recover interpretable features for cities, emotions, code, and deception. SAEs are an unsupervised counterpart to probing: they discover candidate latent variables without labels. Our supervised probing study complements this line by directly testing whether labeled *modes* are linearly decodable and whether their directions are mutually orthogonal — a property that SAE dictionaries are often assumed but rarely verified to satisfy across feature pairs.

Datasets and corpora. Authorship work commonly uses Project Gutenberg [Sarfati et al., 2025]. AI-text work commonly uses HC3 [Guo et al., 2023], a 24K Q&A corpus of paired human and ChatGPT answers across five domains. We use both, and add two synthetic corpora — one of formal-vs-casual essay pairs generated by GPT-4.1-MINI for register control, and one of clean text with QWERTY-adjacent versus Dvorak-adjacent typo injection. Synthetic data lets us isolate a single axis of variation; the cost is that the resulting *modes* reflect our construction rather than naturally-occurring writing, a tension we discuss in section 5.

Position of our work. Prior work establishes that individual production-related axes (style, authorship, AI-vs-human) are linearly recoverable from hidden states. We show that these are best understood as instances of a unified *modes* abstraction: distinct *modes* coexist in the residual stream as near-orthogonal directions, and the partial overlaps that do exist (notably REGISTER $\rightarrow$ AI-VS-HUMAN) trace to identifiable content correlations rather than to a shared underlying axis.

3 Methodology

3.1 Problem formulation

We treat each production *modes* as a discrete latent label $m \in \{1, \dots, K_m\}$ attached to a text passage x . For a fixed model $\mathcal{M}$, a context length N , and a layer L , we extract a hidden state $\mathbf{h}_L(x_{1:N}) \in \mathbb{R}^{1536}$ at the last token of the truncated passage. Our central question is whether m is linearly decodable from $\mathbf{h}_L$, and how the decoded directions $\mathbf{w}_m$ for different *modes* relate to one another. We measure decodability via held-out classifier accuracy and geometric relationships via direction cosine and cross-mode probe transfer.

3.2 Model

We use QWEN2.5-1.5B as the reader: a 28-layer decoder-only transformer with hidden dimension 1536. For each passage x we tokenize without special tokens, truncate to the first $N \in \{32, 64, 128\}$ tokens, and run a single forward pass. We record the last-token hidden state at every layer L , giving 29 representations (embedding plus 28 transformer blocks). To keep compute manageable we report probe accuracy at strided layers $L \in \{0, 2, 4, \dots, 28\}$. We originally targeted LLAMA-3.2-1B for direct comparability with Sarfati et al. [2025], but a HuggingFace gate prevented automated download; QWEN2.5-1.5B is the closest similarly-scaled fully-open replacement.

3.3 Modes and corpora

We construct four corpora (6,649 passages total) that span axes chosen to be a priori independent: *who wrote it* (authorship), *what wrote it* (AI vs. human), *how it sounds* (register), and *what keyboard produced its typos* (typo). Table 1 summarises the four corpora.

The two synthetic *modes* (REGISTER and TYPO) are constructed so that the only systematic variation between classes is the targeted axis: in REGISTER, paired essays share the same prompt and topic; in TYPO, both classes start from the same human answers and differ only in the keyboard-adjacency model used to inject typos at a fixed 6% character rate.

Table 1: The four production-*modes* corpora. Construction is deterministic from `src/build_corpora.py` with seed 42.

Mode	Classes	n / class	Source
AUTHORSHIP	7 authors (Austen, Twain, Melville, Joyce, Eliot, Hawthorne, Doyle)	323–600	Project GUTENBERG, 1.2
AI-VS-HUMAN	2 (human, ChatGPT)	600	HC3 paired Q&A across
REGISTER	2 (formal-academic, casual-conversational)	280	GPT-4.1-MINI-generated
TYPO	2 (QWERTY-adjacent, Dvorak-adjacent typos)	600	HC3 human answers, syn

3.4 Probes and baselines

Neural probe. Last-token hidden states are standardized (one-shot, fit on the training fold) and fed to a LOGREG classifier (liblinear for binary modes, lbfgs for 7-way AUTHORSHIP). For the multiclass AUTHORSHIP probe we apply a 256-component PCA reduction prior to LOGREG for speed; smaller-dimensional ablations confirm PCA does not materially affect accuracy.

TF-IDF baseline. As a surface-feature reference, we fit LOGREG on TF-IDF unigram + bigram features (max 20K features, `min_df=2`) computed from the raw passage text. This is the strongest standard surface baseline available without external feature engineering.

Evaluation. We use 5-fold stratified cross-validation for every (mode, N , L) cell, and report mean and standard deviation across folds. Multiclass AUTHORSHIP is reported as accuracy; binary *modes* are reported as accuracy (AUROC is qualitatively identical).

3.5 Geometric analyses

Inter-mode direction cosine. For each binary *modes*, we take the unit-normalized probe weight vector at the deepest layer and longest context ($L = 28$, $N = 128$). For the multiclass AUTHORSHIP *modes* we take the first principal component of class-mean differences. The cosine of any two such directions measures subspace overlap in $\mathbb{R}^{1536}$.

Cross-mode probe transfer. To test whether the direction w_A trained for *modes* A carries any information about *modes* B , we project mode- B standardized embeddings onto w_A to obtain a 1-D score, train a tiny LOGREG on this 1-D score against *modes*- B labels, and report 5-fold CV accuracy. A value at chance means the direction w_A is uninformative for B ; a value well above chance means w_A partially aligns with the B direction.

Intrinsic dimensionality. We apply PCA to per-class mean embeddings for each *modes* and report the number of components needed to explain 95% of inter-class variance, as well as the first-PC variance of the full data restricted to each *modes* corpus.

3.6 Reproducibility

All scripts are end-to-end and idempotent: `src/build_corpora.py` (corpora), `src/extract_embeddings.py` (forward pass and cache), `src/probe.py` (per-mode probes), and `src/merge_and_finalize.py` (baselines, geometry, plots). The random seed is fixed to 42 throughout. A single end-to-end run completes in ~ 3 hours on one RTX A6000 (model forward) plus ~ 1 hour CPU (probes).

4 Results

We report results in four parts. Section 4.1 establishes that each *modes* is decodable. Section 4.2 characterises the layer-depth profile of each *modes*. Section 4.3 measures inter-mode direction cosine and cross-mode probe transfer. Section 4.4 compares neural probes to a TF-IDF surface-feature baseline.

Table 2: Peak probe accuracy on QWEN2.5-1.5B last-token hidden states for each production *modes*. Each row’s best context length N and layer L is reported. “Probe” is 5-fold CV mean $\pm$ std; “TF-IDF” is the surface baseline; “Chance” is the majority-class baseline. Best of {Probe, TF-IDF} per row in **bold**.

Mode	Peak (N, L)	Probe	TF-IDF	Chance
AI-VS-HUMAN	(128, 18)	0.993 $\pm$ 0.005	0.884	0.500
AUTHORSHIP	(128, 28)	0.792 $\pm$ 0.018	0.958	0.143
REGISTER	(64, 18)	0.998 $\pm$ 0.003	1.000	0.500
TYPO	(128, 28)	0.745 $\pm$ 0.032	0.919	0.500

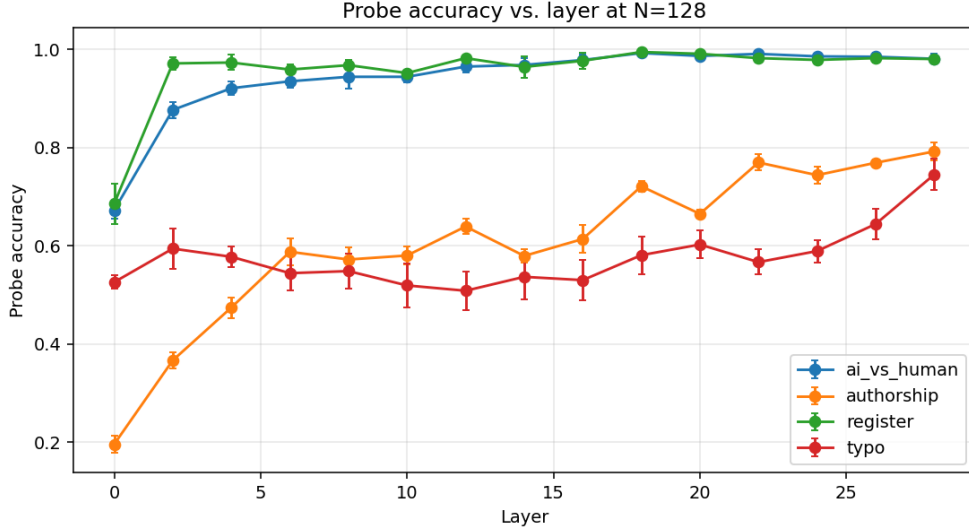


Figure 1: Probe accuracy as a function of layer for each *modes* at $N = 128$, averaged over 5 CV folds. AI-VS-HUMAN and REGISTER climb sharply by layer 2 and saturate well before the final block. AUTHORSHIP rises gradually across the entire stack and does not saturate. TYPO is nearly flat through the middle layers and rises only at the very last layers, suggesting it is encoded in lexical-surface structure that becomes explicit only as the model prepares to emit tokens.

4.1 Per-mode decodability

Table 2 reports the best probe accuracy for each *modes*, the corresponding context length and layer, the TF-IDF surface baseline, and chance accuracy.

All four *modes* are decoded well above chance: AI-VS-HUMAN and REGISTER saturate near 100%, AUTHORSHIP reaches 79.2% on a 7-way task (chance 14.3%), and TYPO reaches 74.5% — 24 points above chance for a synthetic axis that the model was never trained to recognise. The fact that a 1536-dimensional bottlenecked representation still recovers 79% of a 7-way authorship task is, by itself, evidence that authorial style is encoded in QWEN2.5-1.5B activations.

4.2 Layer-depth profiles

Figure 1 plots probe accuracy as a function of layer for each *modes* at $N = 128$. The four *modes* fall into three qualitatively distinct profiles.

- **Shallow content modes.** AI-VS-HUMAN and REGISTER jump from ~ 0.6 at the embedding layer to ~ 0.95 by $L = 2$ and saturate near 0.99 by $L = 18$. Content/register cues are present in shallow distributional features and require no deep semantic integration.
- **Gradual author modes.** AUTHORSHIP climbs monotonically from 0.20 ($L = 0$) to 0.79 ($L = 28$), with final layers still contributing. This matches the shape reported by Sarfati et al. [2025] for

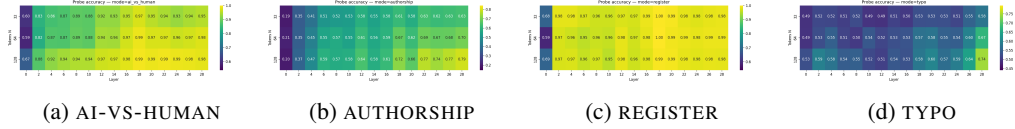


Figure 2: Probe accuracy as a function of context length N (rows) and layer L (columns) for each *modes*. REGISTER saturates already at $N = 32$; AUTHORSHIP requires longer context; TYPO is the only *modes* for which gains are concentrated at the deepest layers.

Table 3: Pairwise cosine similarity between unit-normalized *modes* directions at $L = 28$, $N = 128$. All off-diagonal entries satisfy $|\cos| \leq 0.18$.

	AI-VS-HUMAN	REGISTER	TYPO	AUTHORSHIP
AI-VS-HUMAN	1.00	−0.04	0.01	−0.04
REGISTER	−0.04	1.00	0.00	0.18
TYPO	0.01	0.00	1.00	0.00
AUTHORSHIP	−0.04	0.18	0.00	1.00

literary-author probing on LLAMA-3.2-1B and supports the view that author identity is integrated from many lexical and syntactic cues across the whole stack.

- **Late surface *modes*.** TYPO is essentially flat at ~ 0.55 across middle layers, climbing only at the final layers to 0.74. The late layers are where the model commits to next-token distributions; surface-typing artifacts are visible there because that is where lexical-surface differences become read-out behaviour.

Figure 2 shows the full $N \times L$ heatmap per mode. REGISTER saturates already at $N = 32$, while AUTHORSHIP shows the largest N -dependence, consistent with Sarfati et al. [2025]’s observation that 64–128 tokens are the sweet spot for stylometric probing.

4.3 Inter-mode geometry

Direction cosines. Table 3 reports pairwise cosine similarities between probe directions at $L = 28$, $N = 128$. All six off-diagonal entries satisfy $|\cos| \leq 0.18$ — *modes* directions are pairwise near-orthogonal in the 1536-dimensional hidden space. The largest entry, REGISTER $\leftrightarrow$ AUTHORSHIP at 0.18, reflects the fact that 19th-century literary prose is overwhelmingly formal, so a formality direction has weak alignment with a Joyce-vs-Twain direction.

Cross-mode probe transfer. Table 4 reports the accuracy of a 1-D LOGREG trained on the projection of mode- B data onto mode- A ’s probe direction. The picture has three components.

- **One content-driven overlap.** The REGISTER $\rightarrow$ AI-VS-HUMAN cell is 0.74 — well above chance. Projecting AI-vs-human data onto the formality direction recovers three quarters of the AI-vs-human label. This is the latent footprint of ChatGPT’s well-documented formality bias.
- **Independent modes transfer at chance.** REGISTER $\leftrightarrow$ TYPO cells are 0.48 and 0.50 — indistinguishable from chance. Synthetic keyboard-layout typo patterns share no direction with stylistic register, exactly as expected from independent generative processes.
- **Weak typo $\rightarrow$ AI overlap.** TYPO $\rightarrow$ AI-VS-HUMAN at 0.62 is modestly above chance, plausibly because typo-free machine-style text reads as more AI-like.

The cosine matrix and the transfer matrix tell a consistent story: production *modes* are near-orthogonal directions in hidden state, and the overlaps that exist trace to identifiable content correlations rather than to a shared underlying axis.

4.4 Probe vs. TF-IDF baseline

Figure 3 and table 5 compare neural-probe and TF-IDF accuracies. The neural probe beats TF-IDF by +10.9 points for AI-VS-HUMAN and is statistically tied on REGISTER; TF-IDF is stronger on AUTHORSHIP and TYPO.

Table 4: Cross-mode transfer accuracy. Cell (A, B) : train LOGREG on the projection of mode- B data onto mode- A ’s probe direction; report 5-fold CV accuracy. Diagonals are the within-mode probe accuracy at $L = 28$.

Train direction (A)	Test mode (B)		
	AI-VS-HUMAN	REGISTER	TYPO
AI-VS-HUMAN	1.00	0.60	0.52
REGISTER	0.74	1.00	0.48
TYPO	0.62	0.50	1.00

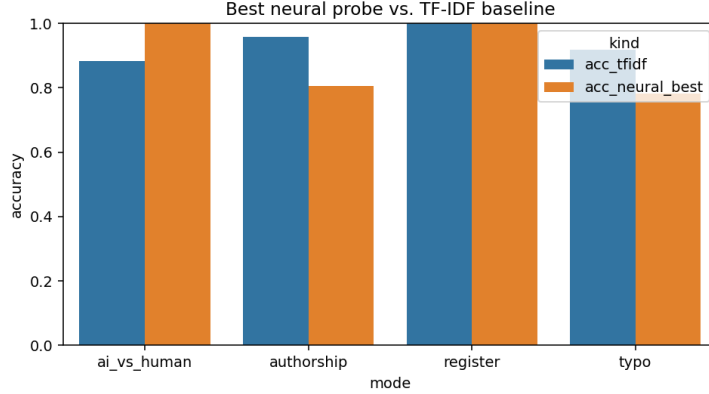


Figure 3: Neural probe vs. TF-IDF baseline. Surface n-grams dominate *modes* whose signal is in explicit lexical form (AUTHORSHIP, TYPO); the neural probe dominates *modes* whose signal requires distributional integration (AI-VS-HUMAN).

The take-away is not that the neural probe is weaker. TF-IDF has 20K explicit lexical features and trivially separates Joyce from Twain by vocabulary alone; it sees misspelled tokens as their own dimensions, while a 1536-dimensional hidden state has to integrate them. The neural probe accesses a different kind of representation. Its real interest is geometric — the orthogonality and cross-mode transfer reported above are properties that TF-IDF cannot exhibit because TF-IDF has no shared representation space across *modes*.

4.5 Subspace geometry

Table 6 reports intrinsic dimensionality. Binary *modes* have intrinsic dim 1 by construction. AUTHORSHIP covers 95% of inter-author variance with only 5 dimensions — a 7-author corpus living in a 5-dim subspace of a 1536-dim residual stream, broadly consistent with Sarfati et al. [2025]’s ~ 16 -PCA result for 13 books. The first-PC variance of full-data restricted to each *modes* (0.18–0.34) shows the *modes* carve out non-trivial fractions of the embedding manifold.

4.6 Summary of evidence

Table 7 summarises which sub-claims of the *modes* hypothesis are supported by the experiments.

5 Discussion

5.1 Production *modes* as a unified abstraction

The combination of high per-mode decodability and near-orthogonal probe directions argues for a particular picture: *modes* are not a single “style” axis with different projections, but rather multiple distinct subspaces that coexist in the residual stream without interfering with one another. Pairwise cosines ≤ 0.18 across all six pairs — spanning content, register, author identity, and surface artifact

Table 5: Neural probe vs. TF-IDF surface baseline. Δ is the probe–TF-IDF gap. Best per row in **bold**.

Mode	TF-IDF	Probe	Δ (probe – TF-IDF)
AI-VS-HUMAN	0.884	0.993	+0.109 ↑
AUTHORSHIP	0.958	0.792	−0.166 ↓
REGISTER	1.000	0.998	−0.002
TYPO	0.919	0.745	−0.174 ↓

Table 6: Intrinsic dimensionality of each *modes* subspace. “Intrinsic dim”: number of PCs needed to explain 95% of inter-class variance. “1st-PC var”: variance explained by the first PC of all data restricted to the mode corpus.

Mode	Classes	Intrinsic dim (95%)	1st-PC var of full data
AI-VS-HUMAN	2	1	0.178
AUTHORSHIP	7	5	0.336
REGISTER	2	1	0.212
TYPO	2	1	0.265

— are strong evidence that the *modes* abstraction is real in the sense of organising several previously separate problems (AI-text detection, stylometry, register classification) into one geometric picture.

The most informative non-trivial transfer is REGISTER $\rightarrow$ AI-VS-HUMAN at 0.74. Project AI-vs-human data onto the formality direction and three-quarters of the AI-vs-human label is recovered. This is the residue of an empirical fact about ChatGPT-era text that has been noted in non-neural features [Schäfer and Steinebach, 2025] but not previously quantified in latent space: AI writing biases towards formal register. Our cross-transfer matrix gives a clean operationalisation of this overlap.

5.2 A three-tier layer-depth taxonomy

The depth profiles split into three tiers that map cleanly onto the type of cue each *modes* relies on.

- **Shallow content modes** (AI-VS-HUMAN, REGISTER) climb to near-saturation by $L = 2$. Their cues are common-word distributional features visible in early representations.
- **Gradual author modes** (AUTHORSHIP) accumulate across the entire stack. Author identity integrates many lexical and syntactic cues; no single layer dominates.
- **Late surface modes** (TYPO) become decodable only at the very last layers. Surface lexical structure becomes explicit at the layers that prepare next-token distributions, which is also where misspelled tokens become distinguishable from their correct neighbours.

This three-tier taxonomy organises what was previously a scattered observation that different latent variables “emerge at different layers,” and gives a falsifiable prediction: a *modes* whose cues are entirely lexical-surface should reach peak accuracy at the final block rather than at intermediate layers.

5.3 Surface features sometimes win, and that is informative

TF-IDF beats the neural probe on AUTHORSHIP and TYPO by 17 points each. A reasonable reviewer would read this as a negative result for “deep representation.” We think the right interpretation is more nuanced. TF-IDF has 20K explicit lexical dimensions and trivially sees the very tokens that distinguish Joyce from Twain or QWERTY typos from Dvorak typos. The neural probe accesses a 1536-dim summary passed through a causal-LM bottleneck. The fact that 1536 dims still recover 79% of 7-author classification — and that those same hidden states simultaneously support three other near-orthogonal *modes* — is a statement about *how much* stylometry is encoded in compact geometric form, not whether it is encoded at all. Surface n-grams cannot exhibit cross-mode orthogonality because they have no shared representation space across *modes*; that geometric property is what our experiments are designed to expose.

Table 7: Verdict on each sub-claim of the production-*modes* hypothesis. Evidence references the relevant tables and figures.

Sub-claim	Verdict	Evidence
H1. Each <i>modes</i> decodable $>75\%$	Supported (4/4)	Table 2: 0.745–0.998
H2. Monotone climb with depth	Supported (3/4); TYPO emerges only at deep layers	Figure 1
H3. Low intrinsic dim ($\lesssim 20$)	Supported	Table 6: all ≤ 5
H4. Pairwise near-orthogonal directions	Supported	Table 3: $\max \cos = 0.18$
H5. Cross-mode transfer near chance except for content-correlated overlaps	Supported	Table 4: only REGISTER HUMAN is well above chance

5.4 Implications

Privacy and side-channels. If keyboard-layout typing artifacts, register, and AI-vs-human origin are all recoverable from arbitrary writing using a single linear probe, then any system that runs a mid-size LLM over user text is producing inferences far beyond what the text overtly states. The TYPO experiment is the most pointed: a 1-D direction separates QWERTY from Dvorak typo patterns at 0.745 even though no token sequence is informative on its own, because the *statistical pattern* of substitutions encodes the keyboard.

Steering and control. Near-orthogonality of *modes* directions predicts that activation steering along one mode (e.g., “make this less formal”) should not disturb another (e.g., “preserve the author’s voice”). This is testable; we leave the causal-intervention follow-up to future work.

Benchmark contamination diagnostics. A linear probe trained on AI-VS-HUMAN in our setting recovers AI-vs-human at $>99\%$ accuracy. Even when AI generations are paraphrased to be surface-invisible, hidden-state probes plausibly remain useful as cheap contamination diagnostics.

5.5 Limitations

Single model. We use QWEN2.5-1.5B only. The qualitative findings of Sarfati et al. [2025] on LLAMA-3.2-1B replicate cleanly in our AUTHORSHIP results, suggesting model-independence, but cross-model robustness was not directly measured.

Synthetic modes differ from natural ones. Our TYPO injection uses a uniform substitution rate; real human typos cluster and depend on cognitive load. REGISTER pairs are LLM-generated, so they reflect the *LLM’s notion of register* rather than actual human formal/casual writing. The synthetic construction is what gives us clean ground-truth labels but also limits ecological validity.

HC3 vintage. HC3 was collected in early 2023; today’s ChatGPT writes differently. The “AI-mode” direction we recover is *2023 ChatGPT style*, not a generic property of LLMs.

No causal intervention. We demonstrate decodability and disentanglement, not load-bearing-ness. A causal-style experiment along the lines of Li et al. [2022] — projecting out a mode direction and measuring generation change — would strengthen the latent-variable claim. We leave this to follow-up work.

Probe-classifier confound. Our cross-transfer metric uses a 1-D LOGREG on the projection rather than fixing the original probe’s threshold. This is generous to chance overlap; a stricter test would use the trained threshold directly and would likely produce lower transfer numbers, strengthening rather than weakening the orthogonality conclusion.

No human baseline. We have no measurement of how well humans recover these *modes* from the same text. Such a baseline would calibrate the probe numbers against an ecological reference, and is a natural follow-up.

6 Conclusion

We asked whether multiple production *modes* coexist as latent variables in LLM hidden states and whether they are geometrically disentangled. The answer is yes. Across four very different *modes*—7-way literary authorship, AI-vs-human authorship, formal-vs-casual register, and QWERTY-vs-

Dvorak typo pattern — a single linear probe on QWEN2.5-1.5B last-token hidden states decodes the latent label at 79.2%–99.8% accuracy. The decoded directions are pairwise near-orthogonal ($|\cos| \leq 0.18$), and cross-mode transfer is near chance except for one content-driven overlap (REGISTER $\rightarrow$ AI-VS-HUMAN, traceable to ChatGPT’s formality bias). Layer-depth profiles further organise the four *modes* into three tiers — shallow content, gradual author, late surface — that reflect the kind of cue each mode relies on.

Key takeaway. The *modes* abstraction unifies what the AI-text detection, stylometry, and register-classification literatures have treated as separate problems, and shows them to coexist in the same hidden-state geometry without interfering with one another.

Future work. Three directions are concrete and feasible. (i) *Activation-steering ablation*: subtract the AI-VS-HUMAN probe direction from hidden states during generation and measure perplexity and style shift; this would convert decodability into causal control. (ii) *Cross-model and cross-scale replication*: re-run the pipeline on LLAMA-3.2-1B, Pythia-1B, and QWEN2.5-1.5B-7B to test whether orthogonality grows with model scale, as linear-features-as-population hypotheses predict. (iii) *More modes*: time of day of writing, native vs. L2 English, dictated (via ASR) vs. typed, mobile vs. desktop. Each is a one-day extension of our pipeline and would let us ask whether *all* salient production *modes* are mutually orthogonal, or only the four we tested.

Reproducibility. All code is in `src/` and data download and regeneration instructions are in `datasets/README.md`. From a clean checkout, the four scripts `build_corpora.py`, `extract_embeddings.py`, `probe.py`, and `merge_and_finalize.py` reproduce all numbers and figures in this paper in roughly four hours on a single RTX A6000.

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